SpectraVault

AI Medical Image Steganography

SEMESTER PROJECT
Digital Image Processing Lab

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CHAPTER 1: ABSTRACT

Medical images contain valuable diagnostic information and are often accompanied by sensitive patient metadata such as identification details, age, gender, and diagnosis. In conventional healthcare systems, this metadata is usually stored separately from the image, increasing the risk of data loss, mismatch, and unauthorized access. To address this issue, this project presents SpectraVault AI, a Digital Image Processing (DIP) framework that securely embeds encrypted patient information within chest X-ray images.

The proposed system combines Artificial Intelligence, Cryptography, and Steganography into a single workflow. DenseNet-121 is used for automated pathology prediction, while AES-256-GCM encryption protects patient metadata before embedding. Three steganographic techniques—Least Significant Bit (LSB), Discrete Cosine Transform (DCT), and Fast Fourier Transform (FFT)—are implemented and compared for secure information hiding.

A web-based application developed using FastAPI and React provides secure signup/login via Supabase Auth, along with embedding and extraction functionalities.

Keywords: Digital Image Processing, Steganography, AES Encryption, LSB, DCT, FFT, Medical Imaging, Chest X-Ray.

CHAPTER 2: INTRODUCTION

2.1 Introduction

Medical imaging has become an essential component of modern healthcare systems. Technologies such as X-ray, CT scan, MRI, and ultrasound assist healthcare professionals in diagnosing diseases and monitoring patient conditions. Along with these images, hospitals maintain a large amount of patient information including identification records, demographic details, and diagnostic reports. Ensuring the secure storage and transmission of this information is a critical requirement in healthcare environments.

Traditionally, patient metadata and medical images are stored separately. Although this approach is widely used, it can lead to data inconsistencies, accidental loss of information, and security vulnerabilities. During data transfer, metadata may become detached from the corresponding image, creating challenges in maintaining accurate patient records.

Steganography provides a potential solution to this problem by embedding information directly into digital images without noticeably affecting their visual quality. When combined with cryptography, steganography not only conceals the existence of information but also protects its confidentiality. This combination creates a secure framework for storing and transmitting sensitive data.

SpectraVault AI explores this concept by integrating encryption and steganography into chest X-ray images. The system encrypts patient metadata using AES-256-GCM and embeds the encrypted information using LSB, DCT, and FFT techniques. In addition, a DenseNet-121 deep learning model is used to analyse chest radiographs and generate pathology labels that can be included in the hidden metadata.

The project demonstrates how Digital Image Processing techniques can be applied to real-world healthcare security challenges while maintaining image quality and data confidentiality.

2.2 Motivation

The increasing use of digital healthcare systems has created a growing need for secure medical data management. Patient records are frequently exchanged between hospitals, laboratories, and healthcare providers, making data protection an important concern.

The primary motivations behind this project include:

- Protecting sensitive patient information from unauthorized access.
- Preventing metadata loss and image-data mismatch.
- Exploring the effectiveness of different steganographic techniques.
- Applying Digital Image Processing concepts in a practical healthcare application.
- Combining Artificial Intelligence, Cryptography, and Steganography within a single framework.

2.3 Scope of the Project

The project focuses on secure embedding of encrypted patient metadata within chest X-ray images. It includes the implementation of AES-256-GCM encryption, LSB, DCT, and FFT steganography techniques, and AI-assisted pathology prediction using DenseNet-121.

The system is intended for educational and research purposes only. It does not aim to replace existing hospital information systems or provide clinical decision-making support.

CHAPTER 3: OBJECTIVES

3.1 Primary Objectives

The primary objectives of SpectraVault AI are:

1. To securely encrypt patient metadata using AES-256-GCM encryption.
2. To implement and compare three steganography techniques: LSB, DCT, and FFT.
3. To embed encrypted information within chest X-ray images while preserving image quality.
4. To accurately recover hidden information through extraction and decryption.
5. To evaluate image quality using PSNR, MSE, and SSIM metrics.
6. To develop a web-based application for embedding and extraction operations.

3.2 Secondary Objectives

The secondary objectives include:

1. Integration of DenseNet-121 for automated pathology prediction.

2. Comparison of spatial-domain and frequency-domain information hiding techniques.
3. Analysis of the relationship between image quality and embedding robustness.
4. Demonstration of practical applications of Digital Image Processing in healthcare security.

CHAPTER 4: THEORETICAL BACKGROUND

4.1 Digital Image Processing

Digital Image Processing (DIP) refers to the use of computational techniques to analyse, manipulate, and improve digital images. It is widely used in medical imaging, computer vision, surveillance systems, and multimedia applications. In this project, DIP techniques are used to transform images, hide information, and evaluate image quality.

4.2 Steganography

Steganography is the process of hiding information within another digital medium such as an image, audio file, or video. Unlike cryptography, which makes information unreadable, steganography conceals the existence of the information itself. The primary goal is to transmit information without attracting attention.

4.3 Least Significant Bit (LSB) Steganography

LSB steganography is one of the simplest and most commonly used information-hiding techniques. It works by replacing the least significant bits of image pixels with message bits. Since these bits contribute very little to overall pixel intensity, changes remain visually imperceptible. LSB provides high image quality and large embedding capacity but is sensitive to compression and image processing operations.

4.4 Discrete Cosine Transform (DCT)

The Discrete Cosine Transform converts image information from the spatial domain into the frequency domain. DCT is widely used in JPEG image compression because it separates important visual information from less significant image details. In this project, encrypted data is embedded into selected DCT coefficients, providing better robustness than traditional LSB methods.

4.5 Fast Fourier Transform (FFT)

The Fast Fourier Transform is an efficient algorithm used to compute the Discrete Fourier Transform. FFT represents image information using frequency components characterized by magnitude and phase. Hidden information can be embedded by modifying selected frequency coefficients while preserving visual quality.

4.6 AES-256-GCM Encryption

AES-256-GCM is a modern authenticated encryption algorithm that provides both confidentiality and integrity. Before embedding, patient metadata is encrypted using a password-based key generated through PBKDF2. This ensures that even if hidden information is extracted, it remains unreadable without the correct password.

4.7 DenseNet-121

DenseNet-121 is a deep learning architecture designed for image classification tasks. In this project, it is used to analyse chest X-ray images and predict possible thoracic diseases. The detected pathology labels can be included in the metadata before encryption and embedding.

4.8 Image Quality Metrics

The performance of the proposed system is evaluated using three standard image quality metrics:

Mean Squared Error (MSE): Measures the difference between the original and stego images. Lower values indicate better quality.

Peak Signal-to-Noise Ratio (PSNR): Measures the quality of the stego image in decibels. Higher values indicate less visible distortion.

Structural Similarity Index Measure (SSIM): Measures structural similarity between two images. Values closer to 1 indicate better preservation of image structure.

CHAPTER 5: SYSTEM DESIGN AND ARCHITECTURE

5.1 System Overview

SpectraVault AI is designed to securely embed encrypted patient metadata into chest X-ray images while maintaining image quality. The system integrates Artificial Intelligence, Cryptography, and Steganography into a single workflow.

The process begins with a chest X-ray image and patient metadata. The metadata is encrypted using AES-256-GCM and then embedded into the image using one of three steganography techniques: LSB, DCT, or FFT. The resulting stego image contains hidden information while appearing visually similar to the original image.

5.2 System Architecture

The system consists of five major modules:

AI Analysis Module

This module uses DenseNet-121 to analyse chest X-ray images and predict possible thoracic diseases. These predictions can be added to the patient metadata.

Encryption Module

Patient metadata is encrypted using AES-256-GCM encryption. A secure key is generated from the user password using PBKDF2.

Steganography Module

The encrypted metadata is embedded into the image using one of the implemented steganographic techniques:

- Least Significant Bit (LSB)

- Discrete Cosine Transform (DCT)
- Fast Fourier Transform (FFT)

Extraction Module

This module extracts hidden information from the stego image and decrypts it using the same password to recover the original metadata.

Authentication Module (Supabase)

This module handles user registration and login. The FastAPI backend connects to Supabase PostgreSQL to create accounts, verify credentials, and issue JWT tokens. Registered users are visible in the Supabase dashboard under Authentication → Users.

5.3 Workflow

The complete workflow of the system is shown below:

Chest X-ray Image → AI Analysis → Metadata Generation → AES Encryption → LSB/DCT/FFT Embedding → Stego Image → Data Extraction → AES Decryption → Recovered Metadata

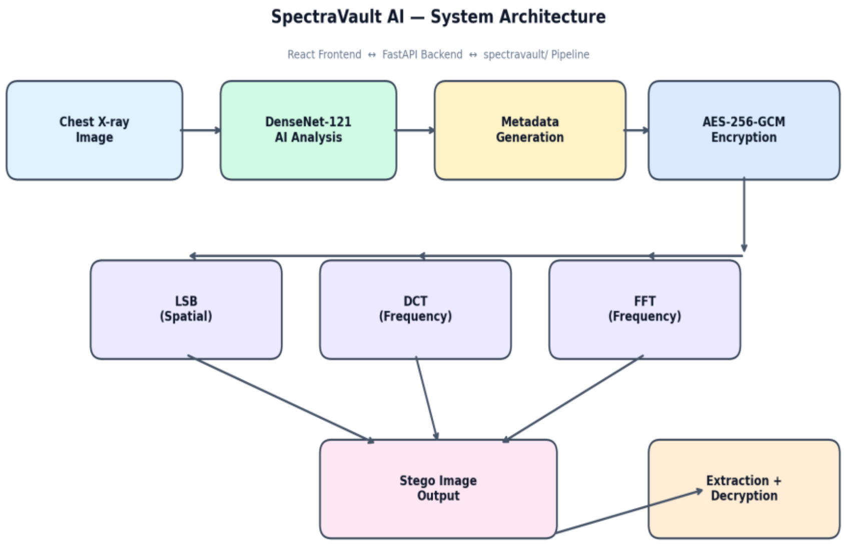


Figure 1: System Architecture Diagram

CHAPTER 6: METHODOLOGY

The proposed methodology consists of four major phases: metadata generation, encryption, data embedding, and extraction.

6.1 Metadata Generation

Patient information including Patient ID, Age, Gender, View Position, and Diagnosis is collected and organized into a structured text format.

Example:

PID: P00001|AGE:40|GEN:M|VIEW:PA|DX: No Finding

If AI analysis is enabled, the detected pathology labels are also included in the metadata.

6.2 Encryption Process

The generated metadata is encrypted using AES-256-GCM. A password-based key is created using PBKDF2, ensuring that the hidden information remains secure.

The encrypted output is converted into ciphertext before the embedding process begins.

6.3 LSB Embedding

In the LSB method, encrypted data bits are embedded into the least significant bits of image pixels. Since only the smallest bit values are modified, visual distortion remains minimal.

6.4 DCT Embedding

In the DCT approach, the image is divided into 8×8 blocks. Each block is transformed into the frequency domain using the Discrete Cosine Transform. Selected coefficients are modified to store encrypted information before converting the block back to the spatial domain.

6.5 FFT Embedding

The FFT method transforms image blocks into frequency components. Encrypted data is embedded by modifying selected frequency magnitudes while preserving phase information. This helps maintain image quality while enabling reliable extraction.

6.6 Data Extraction

During extraction, the hidden bit stream is recovered from the stego image using the selected method. The extracted ciphertext is then decrypted using the correct password to recover the original patient metadata.

CHAPTER 7: IMPLEMENTATION

7.1 Development Environment

The project was implemented using Python for backend processing and React for frontend development. FastAPI was used to create REST APIs for embedding, extraction, and image analysis.

7.2 Backend Implementation

The backend contains the following modules:

- Encryption Module
- LSB Steganography Module
- DCT Steganography Module
- FFT Steganography Module
- AI Analysis Module

- Metrics Evaluation Module

Each module performs a specific task within the overall workflow.

7.3 Frontend Implementation

A React-based web application was developed to provide an interactive user interface. Users can upload images, enter metadata, select a steganography method, and view the resulting outputs.

The frontend includes secure authentication pages and the main application views:

• Login Page

• Signup Page

- • Home Page
- • Embed Page
- • Extract Page
- • Metrics Dashboard

7.4 Dataset

The system was evaluated using the ChestMNIST dataset, which is derived from the NIH ChestX-ray14 dataset. It contains labelled chest radiographs commonly used for medical image classification research.

CHAPTER 8: EXPERIMENTAL SETUP AND RESULTS

8.1 Experimental Setup

The experiments were conducted using Python and the ChestMNIST dataset. For each image, metadata was generated, encrypted, embedded, extracted, and decrypted. The recovered information was compared with the original metadata to verify correctness.

The performance of each method was evaluated using:

- Mean Squared Error (MSE)
- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index Measure (SSIM)
- Extraction Accuracy

8.2 Results

The following results were obtained during evaluation:

Method	PSNR (dB)	SSIM	Accuracy
LSB	71.94	1	100%
DCT	53.51	0.99993	100%

FFT	48.15	0.99972	100%

8.3 Analysis

The results indicate that all three steganography methods successfully recovered the embedded metadata.

LSB achieved the highest PSNR because it introduces very small pixel modifications. DCT and FFT produced slightly lower PSNR values but provided frequency-domain embedding, which is generally considered more robust.

The SSIM values remained close to 1 for all methods, indicating that image structures were preserved. Furthermore, all methods achieved 100% extraction accuracy on the test images, demonstrating the reliability of the proposed framework.

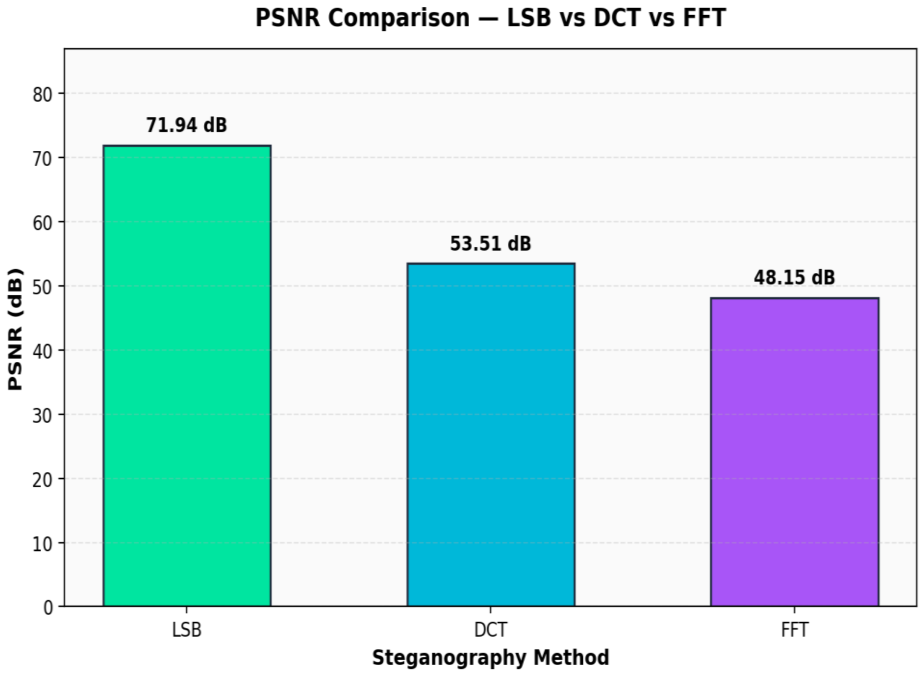


Figure 2: PSNR Comparison of LSB, DCT, and FFT

CHAPTER 9: WEB APPLICATION DEMONSTRATION

To demonstrate the practical implementation of SpectraVault AI, a web-based application was developed using React for the frontend and FastAPI for the backend. The application provides a simple and interactive interface for embedding and extracting encrypted patient metadata from chest X-ray images.

9.1 Home Page

The Home Page provides an overview of the project and explains the workflow of the system. Users can learn about the integration of Artificial Intelligence, Encryption, and Steganography before using the application.

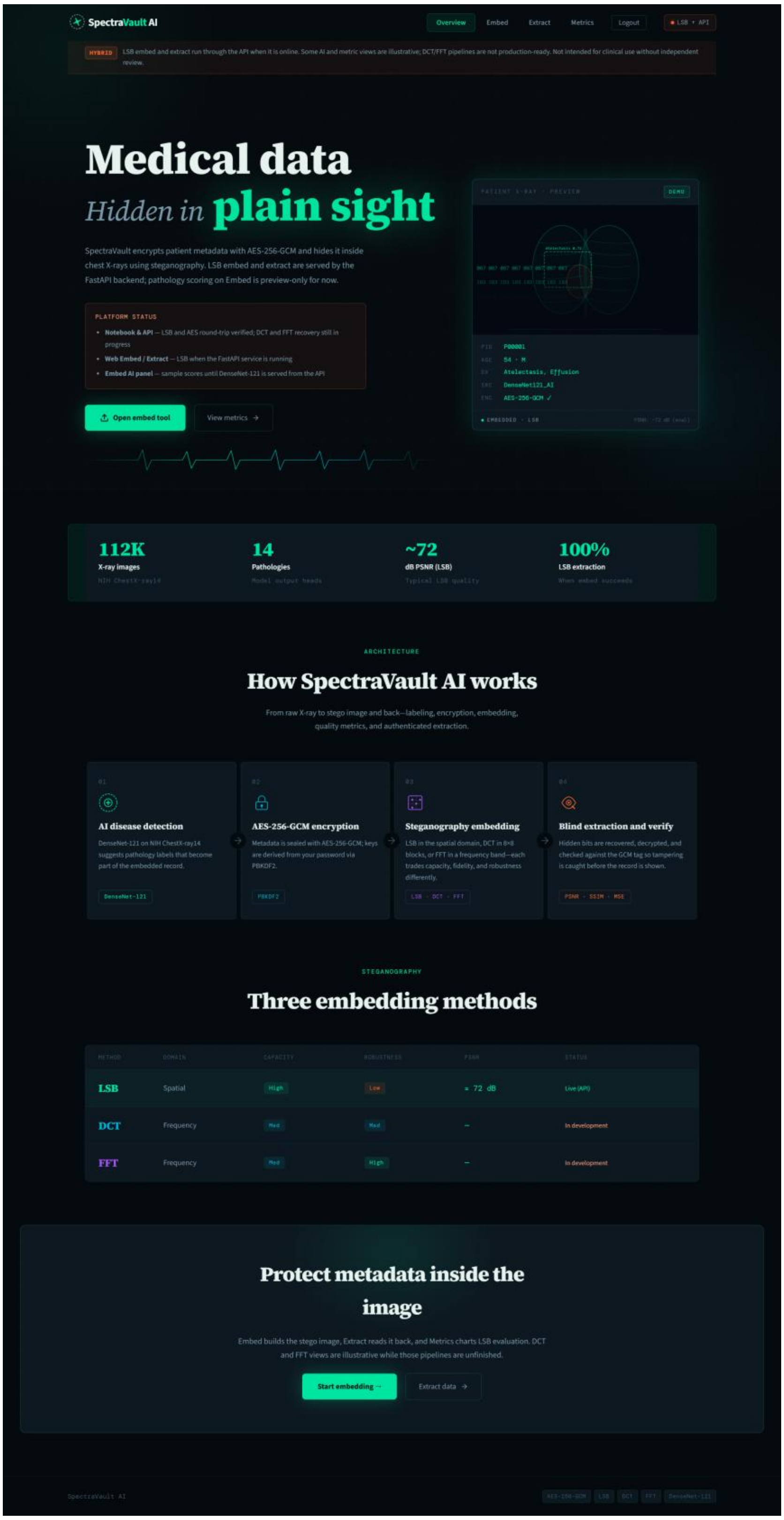


Figure 3: Home Page Interface

9.2 Embed Page

The Embed Page allows users to upload a chest X-ray image and enter patient information such as Patient ID, Age, Gender, and Diagnosis. Users can choose one of the available steganography methods (LSB, DCT, or FFT) and provide an encryption password.

After submission, the metadata is encrypted and embedded into the image. The generated stego image can then be downloaded along with image quality metrics.

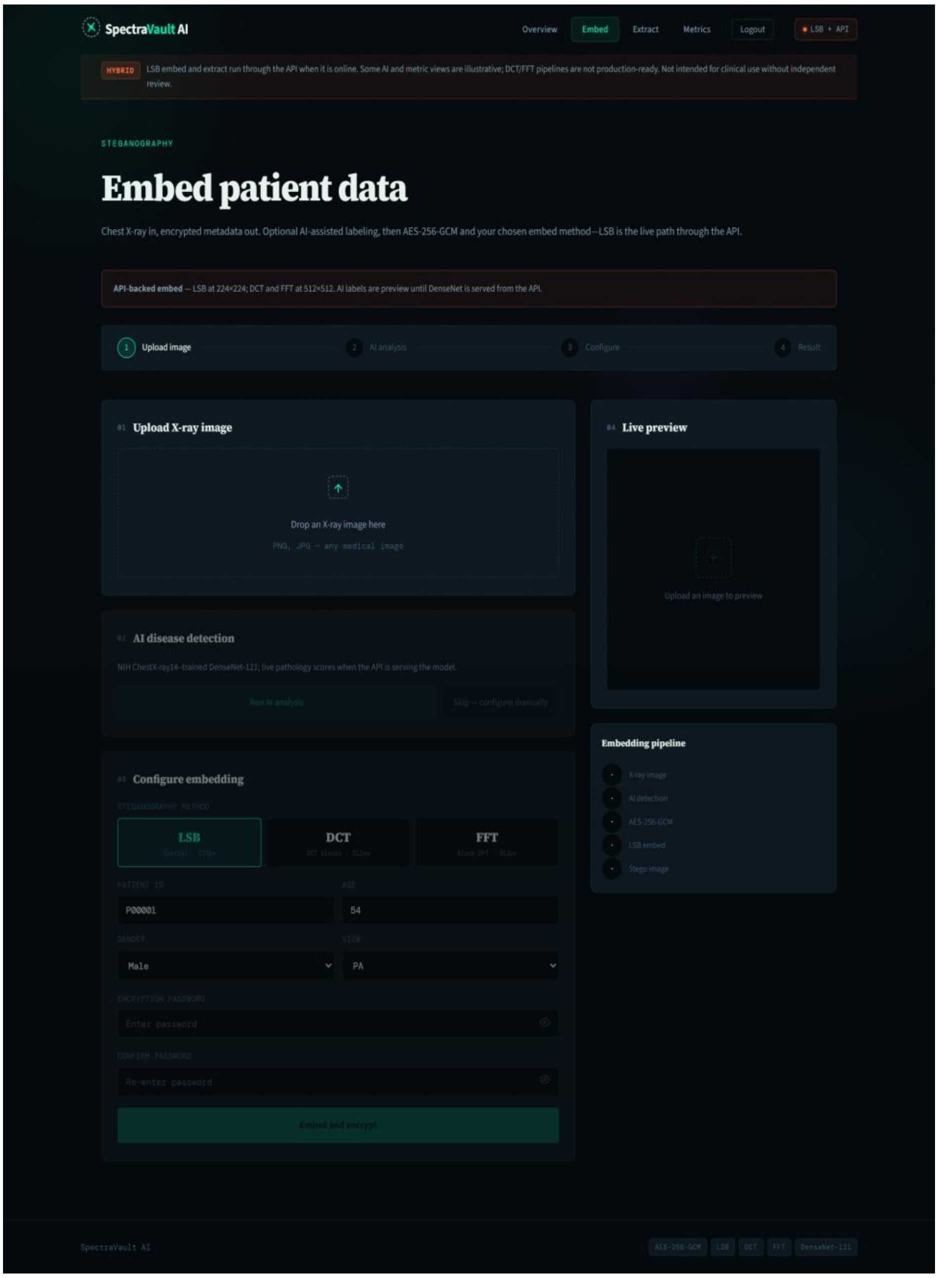
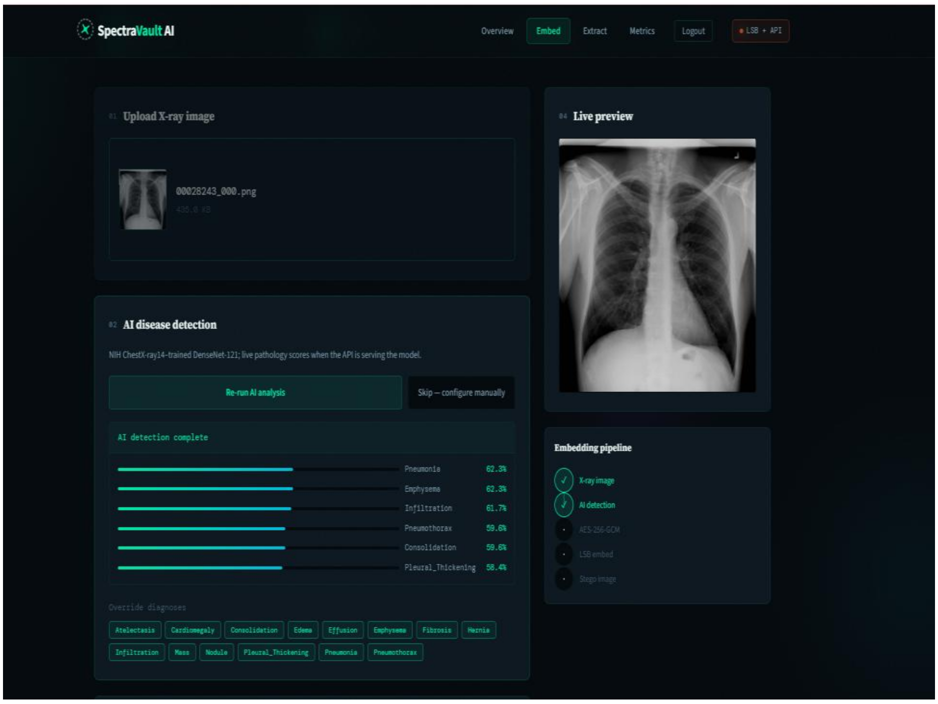
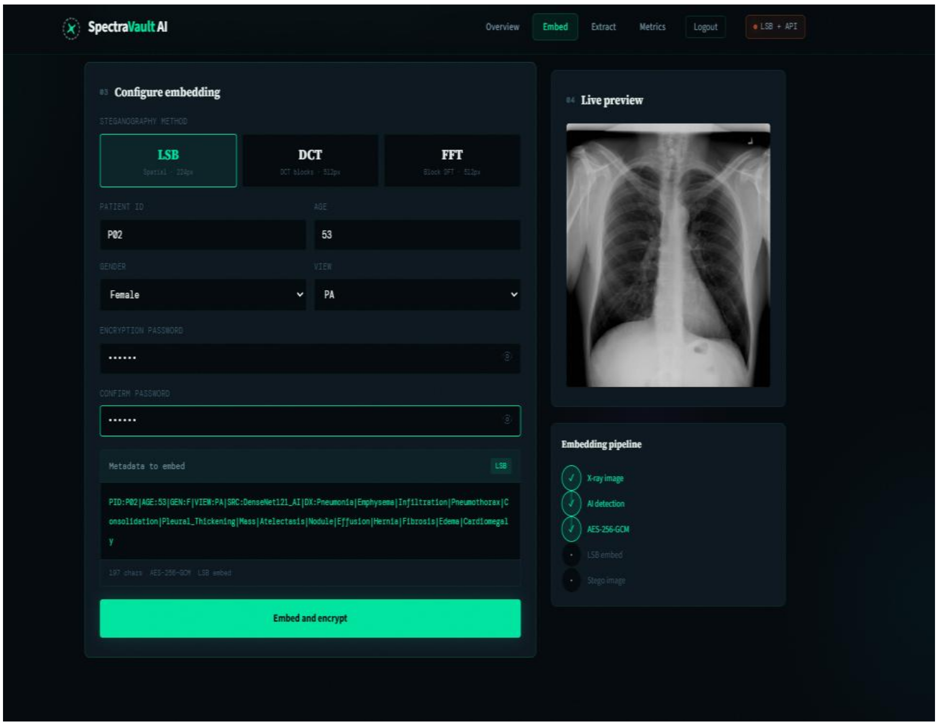
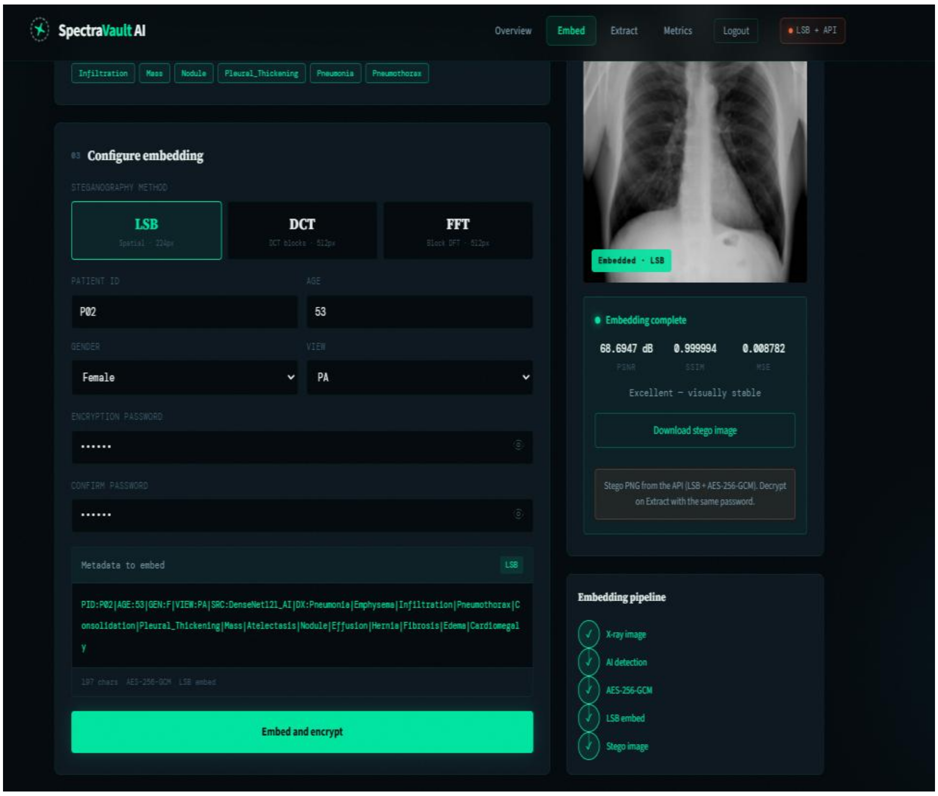


Figure 4: Embed Page


SpectraVault AI
Overview
Embed
Extract
Metrics
Logout
LSB + API
Upload X-ray image
00028243_000.png
Live preview
AI disease detection
NIH ChestX-ray14-trained DenseNet-121; live pathology scores when the API is serving the model.
Re-run AI analysis
Skip — configure manually
AI detection complete
Pneumonia 62.3%
Emphysema 62.3%
Infiltration 61.7%
Pneumothorax 59.6%
Consolidation 59.6%
Pleural_Thickening 58.4%
Embedding pipeline
X-ray image
AI detection
AES-256-GCM
LSB embed
Stego image


SpectraVault AI
Overview
Embed
Extract
Metrics
Logout
LSB + API
Configure embedding
LSB
DCT
FFT
P02
53
Female
PA
Metadata to embed
LSB
PID:P02|AGE:53|GEN:F|VIEW:PA|SRC:DenseNet121_AI|DX:Pneumonia|Emphysema|Infiltration|Pneumothorax|Consolidation|Pleural_Thickening|Mass|Atelectasis|Nodule|Effusion|Hernia|Fibrosis|Edema|Cardiomegaly
Embed and encrypt
Live preview
Embedding pipeline
X-ray image
AI detection
AES-256-GCM
LSB embed
Stego image



9.3 Extract Page

The Extract Page allows users to upload a stego image and enter the correct password. The hidden metadata is extracted, decrypted, and displayed to the user.

This functionality demonstrates the complete recovery process and verifies the reliability of the implemented steganography techniques.

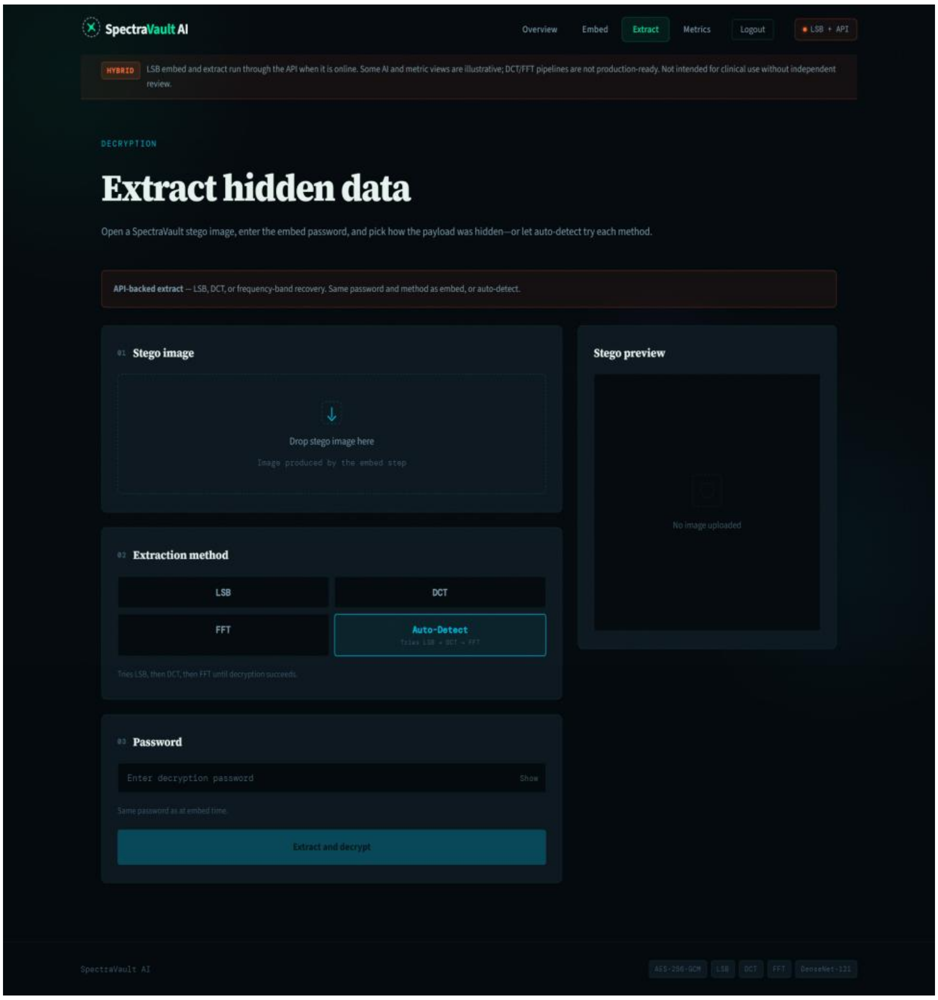
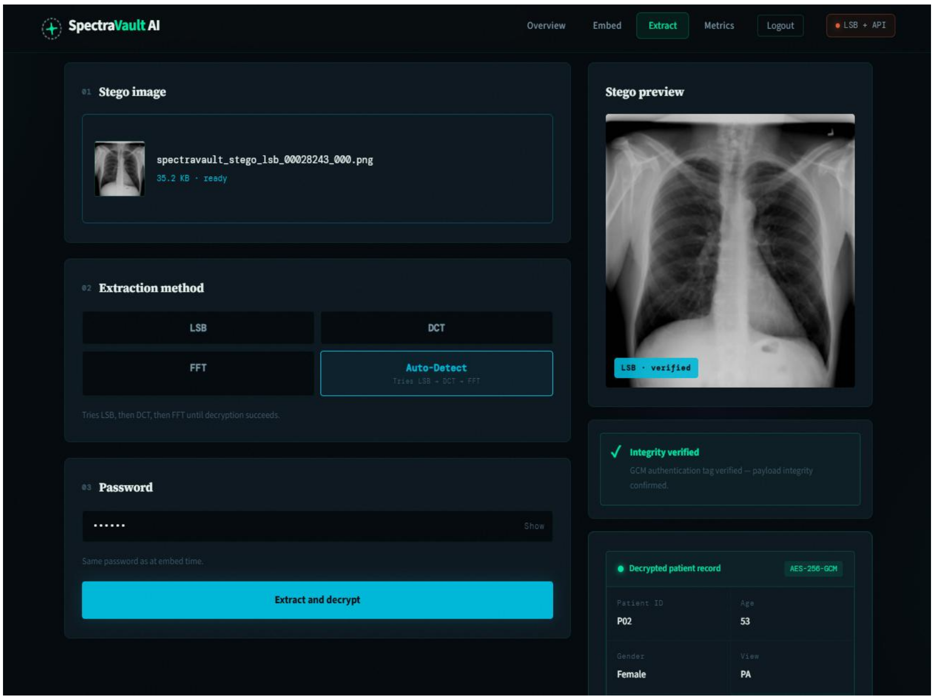
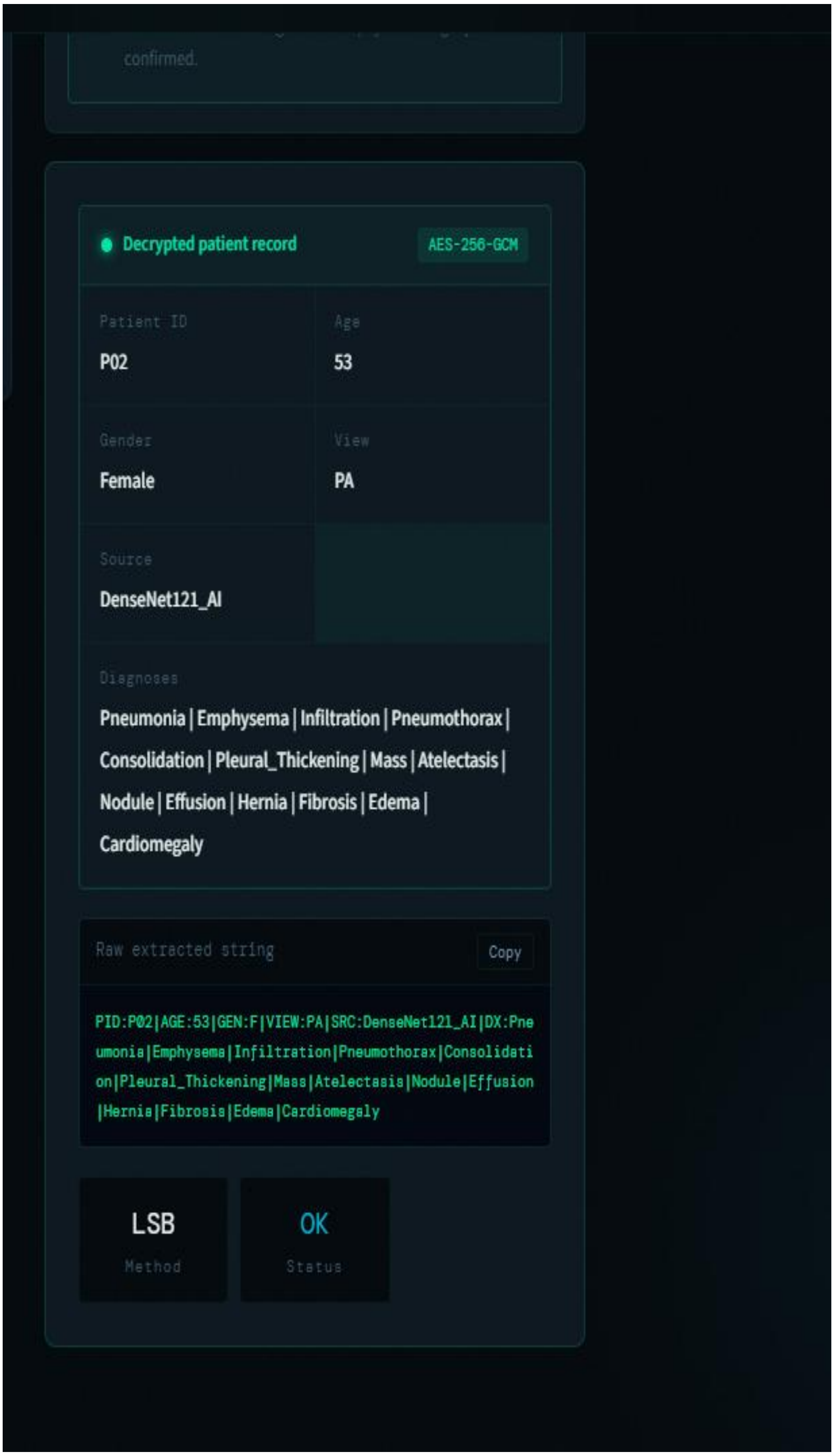


Figure 5: Extract Page




confirmed.
Decrypted patient record
AES-256-GCM
Patient ID
P02
Age
53
Gender
Female
View
PA
Source
DenseNet121_AI
Diagnoses
Pneumonia | Emphysema | Infiltration | Pneumothorax | Consolidation | Pleural_Thickening | Mass | Atelectasis | Nodule | Effusion | Hernia | Fibrosis | Edema | Cardiomegaly
Raw extracted string
Copy
PID:P02|AGE:53|GEN:F|VIEW:PA|SRC:DenseNet121_AI|DX:Pneumonia|Emphysema|Infiltration|Pneumothorax|Consolidation|Pleural_Thickening|Mass|Atelectasis|Nodule|Effusion|Hernia|Fibrosis|Edema|Cardiomegaly
LSB
Method
OK
Status

9.4 Metrics Dashboard

The Metrics Dashboard displays evaluation results generated by the system. It provides visual comparisons of PSNR, MSE, SSIM, and extraction accuracy for all implemented methods.

The dashboard helps users understand the performance differences between LSB, DCT, and FFT steganography.

Figure 6: Metrics Dashboard

9.5 User Authentication and Database Connection

SpectraVault AI implements secure user authentication using Supabase Auth, a cloud-hosted PostgreSQL authentication service. User accounts are stored in Supabase's managed auth.users database table instead of local files. Passwords are hashed by Supabase and are never stored in plaintext on the application server.

The React frontend provides dedicated Signup and Login pages. When a user creates an account, the browser sends a request to the FastAPI backend (POST /api/signup), which registers the user in Supabase using the service role key. On login (POST /api/login), Supabase validates credentials and returns a JWT access token. This token is stored in the browser localStorage and is required to access protected routes such as Home, Embed, Extract, and Metrics.

This database-backed authentication ensures that only registered users can use the embedding and extraction tools. The Supabase dashboard (Authentication -> Users) confirms successful connection to the cloud database and displays all registered users with their email, display name, and unique user ID (UID).

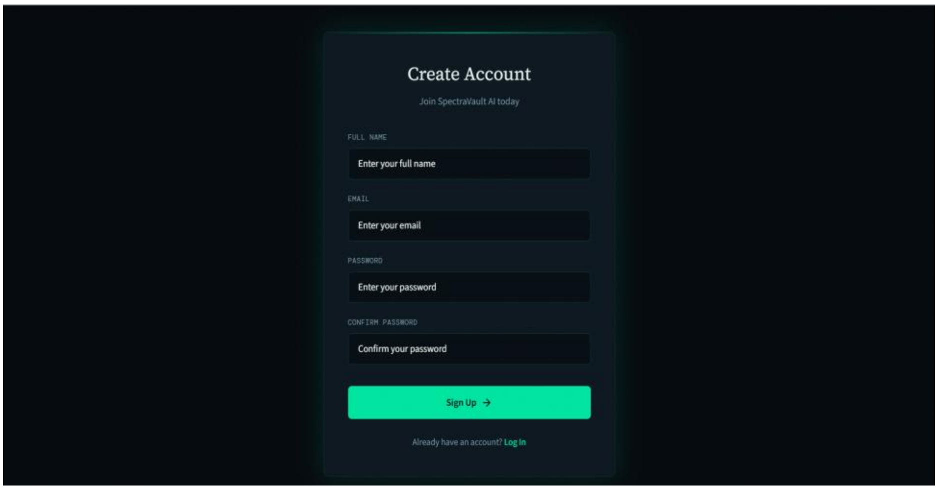


Figure 7: Signup Page

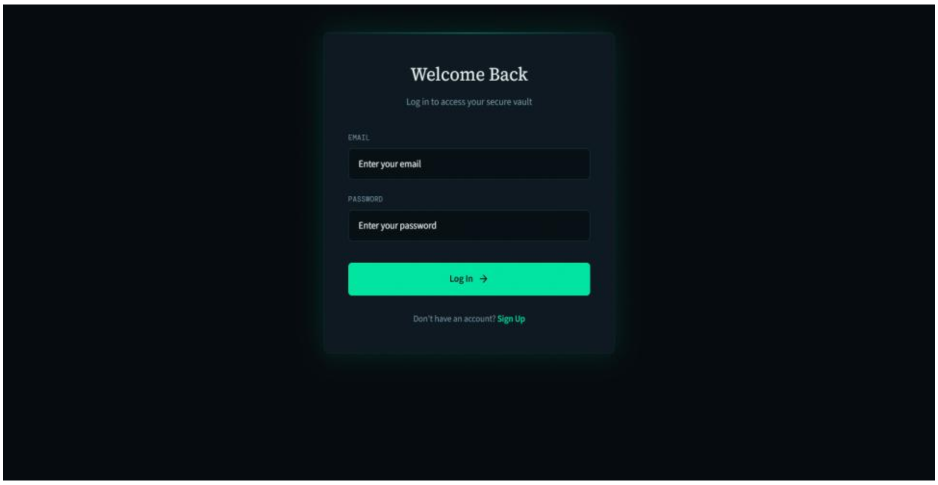


Figure 8: Login Page

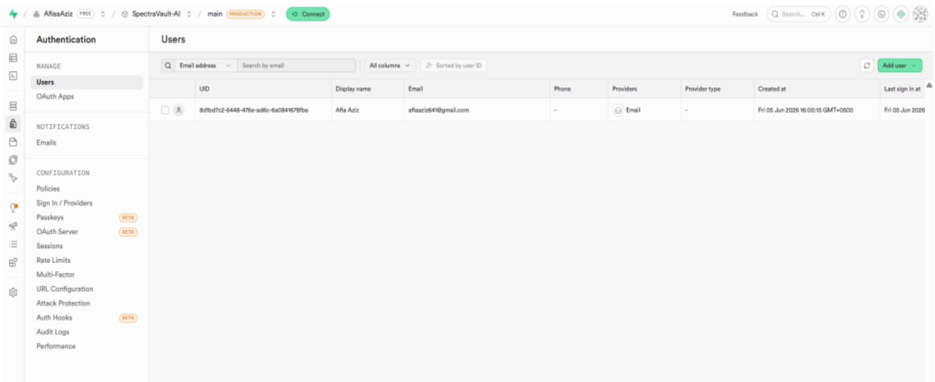


Figure 9: Supabase Authentication Dashboard

CHAPTER 10: DISCUSSION AND LIMITATIONS

10.1 Discussion

The results obtained from the experimental evaluation demonstrate the effectiveness of combining encryption and steganography for secure medical data management.

Among the three techniques, LSB achieved the highest PSNR value, indicating minimal visual distortion. This is because LSB modifies only the least significant bits of image pixels. However, it is more vulnerable to image processing operations such as compression and resizing.

DCT and FFT methods embed information in the frequency domain. Although their PSNR values were lower than LSB, they maintained excellent image quality and offered greater robustness. The high SSIM values obtained for all methods indicate that the structural content of the images remained largely unchanged after embedding.

The integration of AES-256-GCM encryption further enhanced security by ensuring that extracted data remains unreadable without the correct password. This combination of cryptography and steganography provides two layers of protection for sensitive medical information.

10.2 Limitations

Despite its successful implementation, the project has several limitations:

- The system is intended for educational and research purposes only.
- LSB is sensitive to compression, resizing, and image manipulation.
- DCT and FFT methods have lower embedding capacity than LSB.
- Large amounts of metadata require larger image sizes.
- Loss of the encryption password makes metadata recovery impossible.
- The system has not been validated for real clinical deployment.

10.3 Ethical Considerations

The project uses publicly available benchmark datasets rather than real patient records. Any practical deployment in healthcare environments would require compliance with medical privacy regulations and extensive security validation.

The AI component is designed to provide supportive pathology predictions and should not be considered a replacement for professional medical diagnosis.

CHAPTER 11: CONCLUSION AND FUTURE WORK

11.1 Conclusion

SpectraVault AI successfully demonstrates a secure framework for embedding encrypted patient metadata within chest X-ray images. The project integrates Artificial Intelligence, Cryptography, and Digital Image Processing techniques into a unified system.

Three steganography methods—LSB, DCT, and FFT—were implemented and evaluated. Experimental results showed that all methods achieved 100% extraction accuracy while maintaining high image quality. LSB produced the highest PSNR values, whereas DCT and FFT provided frequency-domain alternatives with improved robustness.

The project successfully achieved its objectives and demonstrated the practical application of Digital Image Processing techniques for secure medical information management.

11.2 Future Work

Future enhancements to the project may include:

- Support for DICOM medical image formats.
- Improved robustness against image compression and attacks.
- Integration of error-correcting codes.
- Deep learning-based steganography techniques.
- Production cloud deployment with multi-factor authentication.
- Mobile application development.
- Advanced steganalysis resistance testing.

These improvements can increase the practicality and scalability of the proposed system.

CHAPTER 12: TECHNOLOGY STACK AND PROJECT STRUCTURE

12.1 Technology Stack

The SpectraVault AI system was developed using a modern full-stack architecture:

Backend: Python 3.11, FastAPI, PyTorch, NumPy, Pillow, and the spectravault/ pipeline modules.

Frontend: React 18, Vite, and CSS Modules for a responsive medical-themed user interface.

Security: AES-256-GCM authenticated encryption with PBKDF2 password-based key derivation.

Authentication and Database: Supabase Auth (PostgreSQL auth.users) for secure signup, login, and JWT session management.

Dataset: ChestMNIST (derived from NIH ChestX-ray14) for evaluation and DenseNet-121 training.

12.2 Project Directory Structure

SpectraVault-AI/ contains model.ipynb (notebook pipeline), run_lsb_pipeline.py and run_hybrid_pipeline.py (CLI tools), api/main.py (REST API), spectravault/ (encryption, LSB, DCT, FFT, AI inference), mediguard_outputs/ (metrics and figures), and frontend/ (React web application).

12.3 API Endpoints

POST /api/embed — encrypts metadata and embeds it into an uploaded X-ray using LSB, DCT, or FFT.

POST /api/extract — recovers and decrypts hidden metadata from a stego image.

POST /api/analyze — runs DenseNet-121 pathology prediction on chest X-rays.

GET /api/health — reports API status and available steganography methods.

POST /api/login — authenticates a user and returns a JWT access token.

POST /api/signup — registers a new user account in Supabase Auth.

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